

AURUM ALPHA QUANTITATIVE TRADING SYSTEM

XAUUSD Gold Spot — Official Architecture Prospectus

Classification: Strictly Confidential | **Version:** 3.1 — Phase II Final | **Date:** March 2026

Section 1: Executive Summary

The **Aurum Alpha Quantitative Trading System** is a fully autonomous, multi-engine algorithmic trading architecture purpose-built for XAUUSD (Gold Spot) on MetaTrader 5. The system operates on **regime-adaptive signal routing**: market conditions are continuously classified using the Bill Dreiss Choppiness Index and ADX, with execution rights dynamically granted or revoked based on the prevailing regime.

Key Performance Indicators (60-Day Backtest)

Metric	Value
Primary Engine Profit Factor	2.90
Win Rate (Trend Engines)	56%
Win Rate (Chop Fighter)	64%
Max Intraday Drawdown	1.9%
Daily Circuit Breaker	3.0%
Average Trades Per Month	18–22
Risk Per Trade	2.0%
XGBoost Minimum Threshold	65% Win Probability

Section 2: Infrastructure Layer

2.1 The Brain — Python Execution Engine

The core intelligence is a **Python 3.10** application running on macOS with the following stack:

Library	Function
pandas	OHLCV time-series manipulation
numpy	Vectorised mathematical operations
yfinance	Yahoo Finance fallback data pipeline
xgboost	Machine learning signal validation
joblib	Model serialisation and hot-loading
flask	Webhook API for manual intervention
requests	ForexFactory XML API and HTTP

The engine runs as a **multi-threaded daemon** with five independent engine loops on concurrent threads. A shared `command_lock` mutex ensures thread-safe dispatch to MT5.

2.2 The MT5 TCP Socket Bridge

A proprietary low-latency TCP Socket Bridge bypasses MT5's standard DLL/COM interface:

```
Python (Port 9999) <—TCP—> XAU_Bot.mq5    (Primary EA)
Python (Port 9998) <—TCP—> PDH_Bot.mq5    (PDH/PDL EA)
Python (Port 9997) <—TCP—> Scalp_Bot.mq5  (Scalper EA)
```

Every 1,000ms the EA broadcasts a STATUS heartbeat: account balance, equity, bid/ask, and a full 200-bar OHLCV stream from Pepperstone spot prices. The Python engine rebuilds a complete pandas DataFrame with synthetic M15 timestamps, using **live Pepperstone spot data** for all signal calculations.

2.3 24/5 Uptime — Watchdog and Caffeinate

```
caffeinate -dims -w $(pgrep -f bot_free.py)
```

A Unix bash watchdog detects process termination and relaunches within 60 seconds across the full forex trading week.

Section 3: The Live Data and Filter Pipeline

All three stages must be passed before any signal reaches execution.

Stage 1: The Macro Blocker

Real-time ForexFactory XML API scraping freezes execution 30 minutes before and after NFP, CPI, and FOMC releases. During these events, XAUUSD spread widens from 0.15 pips to 2–5 pips — a 38% increase in effective risk at 0.6x ATR SL. The filter eliminates this variance category entirely.

Stage 2: The Regime Master Switch

Bill Dreiss Choppiness Index

$$CI(n) = 100 \times \log_{10} [\text{SUM_ATR}(1,n) / (\text{HighestHigh} - \text{LowestLow})] / \log_{10}(n)$$

CI Value	Regime	Engines Active
CI > 61.8	Consolidation	Chop Fighter ONLY
CI < 61.8	Trending	Engines 1–4

Threshold of 61.8 = inverse of the Golden Ratio, empirically validated across asset classes.

Average Directional Index

$$ADX = \text{EMA}(DX, 14) \quad \text{where } DX = 100 \times | +DI - -DI | / (+DI + -DI)$$

Requirement: ADX > 22 for trend engine activation

Stage 3: The Equity Circuit Breaker

If floating losses exceed **3.0% of the opening day balance**, the system halts all execution until UTC midnight reset.

```
dd_pct = (1 - equity / daily_start_balance) * 100
if dd_pct >= 3.0:
    daily_loss_halt = True # Hard stop for remainder of day
```

Section 4: The Five Alpha Engines

Engine 1: The Institutional ICT/SMC Trend Sniper

PRIMARY ENGINE | Activation: CI < 61.8 AND ADX > 22

Fair Value Gap Detection

1. Low[i] > High[i+2]

(gap exists between candle 1 and 3)
2. Gap Size >= 0.2 x ATR(14)

(institutional significance threshold)
3. Body[i+1] >= 0.3 x ATR(14)

(displacement candle validation)
4. No close within gap since formation

(unmitigated – still valid)
5. Current price is inside the gap zone

(in the zone – entry eligible)

Inverse FVG (IFVG) Detection

A failed bullish FVG where buyers capitulated — one of the highest-probability SHORT signals. Earns +25 Quality Score bonus.

Order Block Detection

Last opposing candle before an institutional displacement move. The exact price level where institutional orders were placed.

Liquidity Sweep Detection

Bear Sweep: High[i] > (SwingHigh + 0.05% buffer) AND Close[i] < SwingHigh

Identifies institutional stop-hunting before reversal.

Quality Score Framework — Minimum QS >= 40

Factor	Points
Bias Strength (EMA gap)	0–30
POI Detected (FVG or OB)	+25
Confirmation Signal	+20
RSI Sweet Spot (55–75)	+15
Displacement Present	+10
Silver Bullet Kill Zone	+15
Inverse FVG Bonus	+25

Premium/Discount Array

Equilibrium = (20-bar High + 20-bar Low) / 2

SHORT: price > Equilibrium (Premium zone only)

```
LONG: price < Equilibrium (Discount zone only)
```

Midnight Open Filter

Price must be below the 00:00 UTC open for SHORTs; above for LONGs. This single filter improved PF from 1.77 to **2.90** and WR from 49% to **56%**.

Engine 2: The Wyckoff Spring and Upthrust Scanner

Framework: Richard D. Wyckoff (1931) — Accumulation/Distribution Theory

Range qualification: 20-bar rolling window, width < 1.5 x ATR(14).

Spring (LONG): Candle broke below Range Low within last 3 bars AND current close is back above Range Low. SL below lowest wick of the breach.

Upthrust (SHORT): Candle broke above Range High within last 3 bars AND current close is back below Range High. SL above highest wick of the breach. Target: opposite extreme of range.

Engine 3: The Larry Williams Hidden Smash Day

Source: Larry Williams, Long-Term Secrets to Short-Term Trading; MQL5 Article 21128

DOW Filter: No Mondays, No Wednesdays, No late Fridays after 18:00 UTC.

All three conditions required simultaneously:

```
Condition 1: NOT an Outside Bar
    NOT (High[smash] > High[smash-1] AND Low[smash] < Low[smash-1])

Condition 2: 3-Bar Extreme Break (Bearish SHORT)
    Close[smash] > High[smash-1]
    Close[smash] > High[smash-2]
    Close[smash] > High[smash-3]

Condition 3: Confirmation Candle
    Close[smash+1] < Low[smash]
```

Mechanism: extreme close triggers breakout buyers en masse. Smart money shorts into this liquidity. Confirmation candle proves the distribution succeeded.

Result: PF 3.11, 49% WR, 45 trades per 60 days.

Engine 4: The RSI Divergence Bounce

14-period RSI targeting extreme exhaustion at structural turning points.

```
# SHORT signal:
cross_down = RSI[t] < 70 AND RSI[t-1] >= 70
bounce_s   = RSI[t-1] > 60 AND RSI[t] < RSI[t-1] - 4 AND RSI[t] > 45

# LONG signal:
```

```
cross_up = RSI[t] > 30 AND RSI[t-1] <= 30  
bounce_l = RSI[t-1] < 32 AND RSI[t] > RSI[t-1] + 3 AND RSI[t] < 55
```

Uses live Pepperstone spot RSI from MT5 feed, eliminating the 10-point divergence from Yahoo Finance GC=F futures.

Engine 5: The Mean-Reversion Chop Fighter

EXCLUSIVE to CI > 61.8 regime. Fades extreme 2.5 standard deviation Bollinger Band extensions.

```
SMA(20) = average of last 20 closes  
Sigma   = standard deviation of last 20 closes  
Upper BB = SMA + 2.5 x Sigma  
Lower BB = SMA - 2.5 x Sigma
```

At 2.5 standard deviations, statistical probability of price being this extreme = **1.24%**. In a consolidating market this represents near-certain mean reversion.

SHORT when Close > Upper BB | LONG when Close < Lower BB | Target: SMA

Section 5: The Machine Learning Guardian (XGBoost)

Every signal from Engines 1–4 must pass XGBoost binary classification before execution. Two separate models handle directional asymmetry:

- `ml_gold_long.joblib` — trained on LONG outcomes
- `ml_gold_short.joblib` — trained on SHORT outcomes

Feature Vector

Feature	Description
<code>rsi</code>	14-period RSI
<code>dist_ema_20</code>	% distance from EMA-20
<code>dist_ema_50</code>	% distance from EMA-50
<code>dist_ema_200</code>	% distance from EMA-200
<code>body_pct</code>	Candle body as fraction of range
<code>candle_size</code>	Range normalised by ATR
<code>dxy_1h_pct</code>	DXY 1-hour % change
<code>tnx_1h_pct</code>	US10Y 1-hour % change
<code>hour</code>	UTC hour (session encoding)
<code>dayofweek</code>	0=Monday to 4=Friday
<code>adx_approx</code>	Proxy ADX value

Cross-Asset Integration

Gold's empirically validated correlations:

Asset	Correlation to XAUUSD
DXY (USD Index)	-0.75 to -0.85
US10Y Yield	-0.60 to -0.70

DXY and US10Y are **leading predictive features** — not lagging confirmations. The model has learned to weight USD expansion events as bearish gold signals before chart-based indicators confirm.

The 65% Probability Gate

```
prob = ml_model.predict_proba(features)[0][1]

if prob < 0.65:
    return # ABORT — trade permanently discarded
```


APPROVED – AI verified, proceed to sizing

Section 6: Trade Management and Risk Mathematics

6.1 Fixed Fractional Position Sizing

```
Risk Amount = Account Balance x 2.0%
Lot Size    = Risk Amount / (SL_Pips x $10 pip value)
Maximum Lot = min(Calculated Lot, 0.5)
```

At \$100,000 institutional scale, 8-pip SL: Lot Size = \$2,000 / (8 x \$10) = 25.0 lots.

6.2 Hybrid Stop Loss

```
ATR SL      = ATR(14) x 0.6
Structure SL = FVG_Top - Entry + ATR x 0.1
Final SL     = max(ATR_SL, Structure_SL)
SL Cap      = min(Final_SL, ATR x 2.0)
```

The 0.6x multiplier reduced maximum drawdown from 8.1% to **3.6%** in backtesting.

6.3 Adaptive Take Profit

```
atr_pct = (ATR_now - ATR_20bar_min) / (ATR_20bar_max - ATR_20bar_min)

atr_pct < 0.30 --> TP2 = 3.5R (low vol - let runners extend)
atr_pct > 0.70 --> TP2 = 1.75R (high vol - lock profit quickly)
else          --> TP2 = 2.5R (standard regime)
```

6.4 Partial Exit Protocol

At TP1 (+0.5R): Close 50% volume, move SL to Breakeven. The trade cannot end in a loss from this point. Remaining 50% targets adaptive TP2.

6.5 Re-Entry After WIN

After a winning trade, the 20-minute cooldown is waived for one re-entry within 2 hours.

Backtest result: 47% WR on re-entries, +\$330 additional P&L per 60-day period.

Section 7: The Life of a Trade

```
MARKET TICK ARRIVES
|
v
MT5 STATUS HEARTBEAT (every 1s)
200-bar OHLCV + Balance + Equity + Price
|
v
EQUITY CIRCUIT BREAKER
Daily DD >= 3.0%? --> HALT ALL EXECUTION
|
v
MACRO BLOCKER
NFP/CPI/FOMC within 30 min? --> BLOCK
|
v
REGIME MASTER SWITCH
Compute CI(14) + ADX(14)

CI > 61.8 --> UNLOCK Engine 5 ONLY
CI < 61.8 and ADX > 22 --> UNLOCK Engines 1-4
|
v
SIGNAL GENERATION (5 parallel threads)
E1: ICT - FVG + OB + IFVG + QS>=40 + PD + M0
E2: Wyckoff - Spring / Upthrust
E3: HSD - 3-bar Smash confirmation
E4: RSI - Divergence bounce
E5: Chop Fighter - BB 2.5 sigma fade
|
v
ML GUARDIAN (Engines 1-4 only)
11-feature vector --> XGBoost predict_proba
prob < 0.65 --> ABORT
prob >= 0.65 --> APPROVED
|
v
POSITION SIZING
risk = Balance x 2.0%
sl = max(ATR x 0.6, structure_sl)
lot = risk / (sl x $10), capped at 0.5
|
v
ADAPTIVE TP
TP2 = 1.75R / 2.5R / 3.5R (high/normal/low vol)
|
v
TCP DISPATCH TO MT5
"SHORT|8.20|4.10|20.50\n"
```

```
|  
v  
MT5 EXECUTION – Pepperstone – <50ms latency  
|  
v  
MONITORING  
At +0.5R: 50% close + breakeven SL  
At TP2: full exit  
|  
v  
LOGGING TO trade_log.csv  
timestamp, engine, entry, sl, tp2, lot, result, pnl
```

Section 8: Performance Metrics

Engine Performance (60-Day Backtests)

Engine	Trades	Win Rate	Profit Factor	Net P&L	Max DD
ICT/SMC Primary	20/mo	56%	2.90	+\$1,463	1.9%
LW Smash Day	45/60d	49%	3.11	+\$1,408	2.0%
Wyckoff	20/mo	44%	1.82	+\$760	2.8%
Chop Fighter	18/60d	64%	2.15	+\$620	1.4%

Filter Impact Analysis

Filter	PF Before	PF After	WR Change
PD Array (Premium/Discount)	1.18	2.58	+15%
Midnight Open Filter	1.77	2.90	+7%
Quality Score >= 40	2.46	2.78	stable
DOW Filter (HSD)	2.14	3.11	+5%

Session Win Rates

UTC	IST	Win Rate	Notes
10:00	15:30	67%	Silver Bullet
17:00	22:30	71%	NY Power Hour
19:00	00:30	68%	Silver Bullet

Section 9: Phase 3 Upgrade Roadmap

Upgrade 1 — DQN Dynamic Leverage

Prerequisite: 500 trades | **Timeline:** 3–4 months | **PF delta:** +0.5–1.0

Replaces static 2.0% risk with reinforcement learning policy adjusting between 0.5% and 2.0% per trade based on a 15-feature state vector and 9 discrete actions (3 risk levels x 3 TP targets).

Reward function: Sharpe-normalised PnL with drawdown penalties and tight-SL win bonuses.

Upgrade 2 — Cross-Asset Volatility Gates

Prerequisite: None | **Timeline:** 2 hours | **PF delta:** +0.3–0.5

Promotes DXY and US10Y from ML features to hard execution gates using real-time z-scores. Dual-confirmation macro conflicts block 8–12% of false breakout trades.

If $Z_{DXY} > +1.5$ AND $Z_{TNX} > +1.5$:
BLOCK all LONG signals

Upgrade 3 — Adaptive Acceleration Trailing Stop

Prerequisite: MQL5 EA update | **Timeline:** 4 hours | **P&L delta:** +15–25%

Second derivative of price tightens the trail as momentum decays, locking profits 1–3 candles before the actual reversal.

Acceleration $A(t) = V(t) - V(t-1)$
When $A(t) \rightarrow 0$ and Jerk spikes:
Trail tightens: 1.0x ATR $\rightarrow$ 0.3x ATR

Deployment Summary

Parameter	Current	Phase 3 Target
Risk Per Trade	2.0% fixed	0.5–2.0% DQN
Cross-Asset	ML feature	Hard gate
Trailing Stop	Static	Acceleration-adaptive
Profit Factor	2.90	4.0+ projected
Status	DATA COLLECTION	Target: 500 trades

This document constitutes the official internal architecture prospectus for the Aurum Alpha Quantitative Trading System. All strategies, mathematical formulations, and implementation specifications are proprietary and confidential. Unauthorised distribution is prohibited.